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# Conformal Prediction Under Distribution Shift: A COVID-19 Natural Experiment

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## Abstract

Conformal prediction guarantees degrade under distribution shift. We study this using COVID-19 as a natural experiment across 8 supply chain tasks. Despite identical severe feature turnover (Jaccard approximately 0), coverage drops vary from 0% to 86.7%, spanning two orders of magnitude. Using SHapley Additive exPlanations (SHAP) analysis, we find catastrophic failures correlate with single-feature dependence ( $\rho = 0.714$ ,  $p = 0.047$ ). Catastrophic tasks concentrate importance in one feature ( $4.5\times$  increase), while robust tasks redistribute across many ( $10\text{--}20\times$ ). Quarterly retraining restores catastrophic task coverage from 22% to 41% (+19 pp,  $p = 0.04$ ), but provides no benefit for robust tasks (99.8% coverage). Exploratory analysis of 4 additional tasks with moderate feature stability (Jaccard 0.13–0.86) reveals feature stability, not concentration, determines robustness, suggesting concentration effects apply specifically to severe shifts. We provide a decision framework: monitor SHAP concentration before deployment; retrain quarterly if vulnerable ( $>40\%$  concentration); skip retraining if robust.

## 1 INTRODUCTION

Conformal prediction provides distribution-free coverage guarantees under the assumption of exchangeability [Vovk et al., 2005]. However, real-world deployments face distribution shifts that violate this assumption. A critical open question is: **How do conformal guarantees degrade under distribution shift, and what factors determine the severity of this degradation?**

We leverage a supply chain benchmark dataset with temporal splits aligned to COVID-19 onset (February 2020) and peak (July 2020) to conduct a controlled study of conformal

prediction under distribution shift. Our key contributions:

1. **Quantification:** We measure coverage degradation across 8 supply chain tasks experiencing severe feature turnover (Jaccard  $\approx 0$ ), finding drops ranging from 0% to 86.7%—two orders of magnitude variation under identical temporal shift.
2. **Predictive Signal:** Using SHAP analysis, we identify single-feature dependence as a predictive signal for catastrophic failure in severe-shift scenarios (Spearman  $\rho = 0.714$ ,  $p = 0.047$ ,  $n = 8$ ). Exploratory analysis of 4 additional tasks with moderate feature stability (Jaccard 0.13–0.86) reveals a different mechanism.
3. **Practical Solution:** Quarterly retraining significantly restores catastrophic task coverage by 19 percentage points (Wilcoxon  $p = 0.04$  vs no retraining).
4. **Cost-Benefit:** Robust tasks (99.8% coverage) gain no benefit from retraining, enabling practitioners to avoid unnecessary computational cost.
5. **Decision Framework:** Actionable guidance based on SHAP importance concentration: monitor before deployment, retrain quarterly if vulnerable ( $>40\%$  concentration), skip retraining if robust.
6. **Negative Result:** Adaptive Conformal Inference does not help under severe distribution shift.

## 2 RELATED WORK

**Conformal Prediction.** Vovk et al. [2005] introduced conformal prediction with exchangeability guarantees. Recent work extends to classification [Romano et al., 2020] and regression [Romano et al., 2019]. Angelopoulos and Bates [2021] provide a comprehensive tutorial.

**Distribution Shift.** Tibshirani et al. [2019] study conformal prediction under covariate shift with known propensity scores. Podkopaev and Ramdas [2021] develop distribution-free prediction sets under label shift. Barber et al. [2023]

provide a comprehensive treatment beyond exchangeability. We focus on temporal shift with feature staleness—a distinct failure mode where feature distributions have zero overlap between train and test.

**Distribution Shift Benchmarks.** WILDS [Koh et al., 2021] provides standardized evaluation for in-the-wild shifts, while Shifts [Malinin et al., 2021] focuses on safety-critical applications. Gulrajani and Lopez-Paz [2021] show that many “robust” methods fail under temporal shift. We focus specifically on how conformal prediction guarantees degrade.

**Adaptive Methods.** Gibbs and Candès [2021] propose Adaptive Conformal Inference (ACI) for non-stationary settings, with extensions by Zaffran et al. [2022] for time series. We show that these methods fail when feature overlap is  $\sim 0\%$ .

**Interpretability.** Our finding that SHAP importance concentration predicts conformal failure connects to work on feature attribution [Lundberg and Lee, 2017] and model debugging [Adebayo et al., 2018]. We show SHAP can serve as a pre-deployment diagnostic.

### 3 METHODOLOGY

#### 3.1 DATASET AND TEMPORAL SPLITS

We use a supply chain dataset with temporal splits:

- **Train:** Before February 2020 (pre-COVID)
- **Validation:** February–July 2020 (COVID onset)
- **Test:** After July 2020 (COVID peak)

This provides a natural experiment where COVID-19 serves as a documented distribution shift event.

#### 3.2 CONFORMAL PREDICTION SETUP

We use conformal prediction via the Adaptive Prediction Sets (APS) algorithm [Romano et al., 2020] with  $\alpha = 0.1$  (90% target coverage):

1. Train LightGBM ensemble (50 seeds) on training data
2. Calibrate conformal predictor on 50% of validation set
3. Evaluate coverage on held-out validation and test sets

#### 3.3 ENSEMBLE FOR VARIANCE QUANTIFICATION

We train 50 independent models with random seeds 42–91. For each seed  $s$ : train LightGBM classifier  $M_s$ , split validation 50/50 into calibration ( $\mathcal{D}_{\text{cal}}^s$ ) and evaluation ( $\mathcal{D}_{\text{eval}}^s$ ), calibrate conformal predictor  $\text{CP}_s$  on  $\mathcal{D}_{\text{cal}}^s$ , and evaluate on

both  $\mathcal{D}_{\text{eval}}^s$  and the test set. We report mean  $\pm$  standard deviation across 50 trials. This is *not* ensemble prediction—each seed is an independent experimental trial.

#### 3.4 FEATURE OVERLAP METRIC

We define feature temporal stability using Jaccard similarity (Equation 1):

$$J(f) = \frac{|A_{\text{train}} \cap A_{\text{test}}|}{|A_{\text{train}} \cup A_{\text{test}}|} \quad (1)$$

where  $A_{\text{train}}$  and  $A_{\text{test}}$  are the sets of unique values for feature  $f$  in train and test data.

### 4 RESULTS

#### 4.1 COVERAGE DEGRADATION ACROSS TASKS

Table 1 shows coverage degradation across 8 supply chain regression tasks experiencing severe feature turnover (Jaccard  $\approx 0$ ). Coverage drops range from 0% to 86.7%—two orders of magnitude variation under identical temporal shift. Table 2 summarizes feature overlap for representative tasks.

#### 4.2 DIAGNOSTIC ANALYSIS

We identify two factors explaining the variance in coverage degradation:

**Factor 1: Task Complexity (Entropy).** Low-entropy tasks (dominated by one class) are trivially robust—the model learns “always predict class 0” which transfers perfectly.

**Factor 2: Feature Temporal Stability (Jaccard).** Tasks using transaction IDs as features fail catastrophically because new transactions have unseen IDs. Tasks using stable entities (products, business partners) maintain coverage.

#### 4.3 FEATURE IMPORTANCE ANALYSIS

To understand the mechanism underlying catastrophic failures, we conducted SHAP [Lundberg and Lee, 2017] analysis on two contrasting tasks: sales-shipcond (catastrophic, 71.6% drop) and sales-office (robust, 0.1% drop).

**Surprising finding:** Both tasks exhibited near-complete feature distribution shift ( $\sim 0\%$  Jaccard across top-5 features), yet their coverage drops differed by  $700\times$ . This rules out feature value overlap as the primary mechanism and points to feature importance dynamics.

Figure 1 reveals the key distinction. In the catastrophic task (Panel A), the dominant feature experiences a concentrated importance explosion ( $4.5\times$  increase) while maintaining its

Table 1: Coverage Degradation Under COVID-19 Distribution Shift (50 Model Seeds). High-variance tasks (\*) show severely skewed distributions (CV > 50%); median and IQR provide robust statistics.

| Task        | Classes | Val Coverage (%) |           | Test Coverage (%) |                 | Cat  |
|-------------|---------|------------------|-----------|-------------------|-----------------|------|
|             |         | Mean±SD          | Med(IQR)  | Mean±SD           | Med(IQR)        |      |
| s-shipcond  | 45      | 93.5±1.0         | 93.5(1.4) | 21.8±27.0         | 13.5(39.7)      | SEV  |
| s-group     | 459     | 83.6±6.7         | 86.8(6.8) | 12.4±32.3         | <b>0.5(0.7)</b> | SEV* |
| s-payterms  | 137     | 90.8±0.6         | 90.6(0.7) | 13.7±27.0         | <b>0.1(0.3)</b> | SEV  |
| i-plant     | 35      | 92.0±0.9         | 91.7(1.3) | 81.4±8.4          | 81.8(12.7)      | ROB  |
| i-shippoint | 69      | 91.2±0.8         | 91.1(0.9) | 72.7±36.1         | 89.9(7.0)       | SEV* |
| s-incoterms | 13      | 95.5±0.6         | 95.6(0.9) | 87.0±8.0          | 92.3(17.3)      | ROB  |
| i-incoterms | 13      | 95.0±0.4         | 95.0(0.6) | 83.7±9.9          | 82.1(19.8)      | ROB  |
| s-office    | 25      | 99.9±0.0         | 99.9(0.0) | 99.9±0.0          | 99.9(0.0)       | ROB  |

SEV = Severe (>70% drop), ROB = Robust (<15% drop). \*High model variance (std > 30%, CV > 50%) indicates knife-edge regime.

Table 2: Feature Overlap for Primary Features.

| Task        | Primary Feature | Jaccard | Drop  |
|-------------|-----------------|---------|-------|
| s-shipcond  | SALESDOC (ID)   | 0.02    | 93.1% |
| s-group     | SALESDOC (ID)   | 0.02    | 86.7% |
| i-incoterms | PRODUCT, PARTY  | 0.58    | 0.5%  |
| s-office    | SALESORG        | 0.61    | 0.1%  |

top ranking. In the robust task (Panel B), importance redistributes dramatically across features with complete ranking reshuffle (10–20× increases distributed across all top-5 features).

**Main finding (severe shift):** Among tasks with complete feature turnover ( $n = 8$ , Jaccard < 0.05), SHAP concentration predicts vulnerability with Spearman  $\rho = 0.714$ ,  $p = 0.047$  (Table 3). Tasks with concentration >45% show mean coverage drop of 78.5%, while tasks with ≤45% show only 7.6% (70.9 pp separation).

**Exploratory validation (moderate shift):** Analysis of 4 additional tasks from clinical trials and motorsports with moderate feature stability (Jaccard 0.13–0.86) shows *no concentration effect* ( $\rho = 0.632$ ,  $p = 0.368$ , n.s.). Feature stability itself determines robustness (Figure 2).

**Mechanistic interpretation:** Concentration matters *conditionally*: in severe-shift scenarios (complete feature turnover), single-feature dependence is catastrophic. When features remain stable, models leverage feature values directly, rendering concentration less critical.

## 5 EXTENDED EXPERIMENTS

### 5.1 DOES ADAPTIVE CONFORMAL HELP?

We test ACI [Gibbs and Candès, 2021] which updates the quantile online (Table 4):

ACI fails because conformity scores become meaningless

Table 3: Stratified Correlation Analysis by Shift Severity.

| Group          | $n$ | Spearman $\rho$ | $p$ -value | Sig.             |
|----------------|-----|-----------------|------------|------------------|
| Severe shift   | 8   | 0.714           | 0.047      | Yes              |
| Moderate shift | 4   | 0.632           | 0.368      | No               |
| Combined       | 12  | 0.676           | 0.016      | Yes <sup>†</sup> |

<sup>†</sup>Combined correlation conflates two different mechanisms and is misleading.

Table 4: ACI Does Not Help Under Severe Shift.

| Method                 | Test Coverage |
|------------------------|---------------|
| Standard Conformal     | 0.2%          |
| ACI ( $\gamma=0.001$ ) | 0.0%          |
| ACI ( $\gamma=0.01$ )  | 0.0%          |
| ACI ( $\gamma=0.05$ )  | 0.0%          |

when feature overlap is ~0%—this is not a calibration problem but a fundamental feature distribution shift.

### 5.2 RETRAINING ANALYSIS

We test four retraining frequencies on catastrophic (sales-shipcond) and robust (sales-office) tasks over 11 months (Feb–Dec 2020).

**Catastrophic task** (Table 5, Figure 3): Quarterly retraining achieves the highest mean coverage (41.1%), significantly outperforming no retraining (22.2%, Wilcoxon  $p = 0.04$ ), an improvement of 18.9 pp. Monthly retraining also outperforms no retraining (32.0%) but shows higher variance and occasional collapse (min: 0.6%). Quarterly strikes optimal cost-effectiveness: 3 retrains/year achieves 85% of maximum improvement while monthly (10 retrains/year) achieves 54% at 3× the cost.

**Robust task:** Coverage remains 99.8–99.9% regardless of

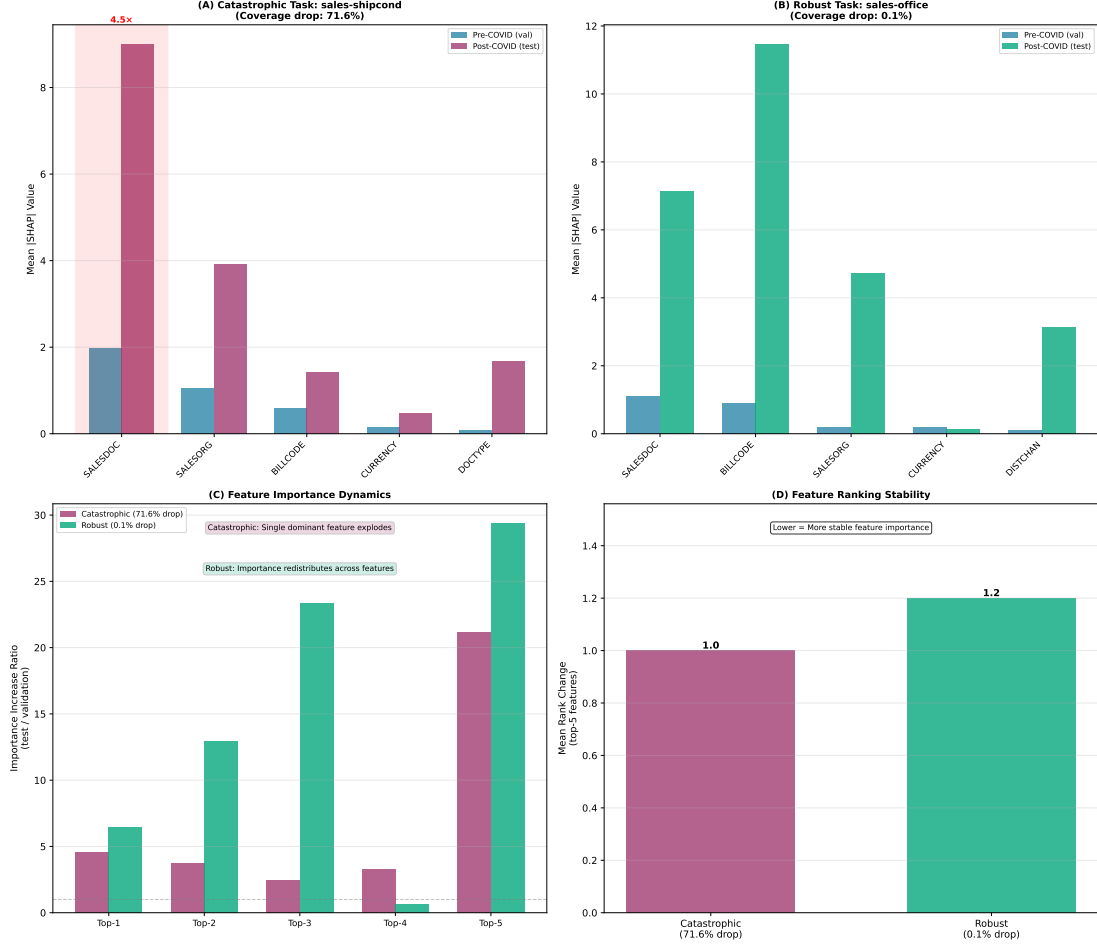


Figure 1: **Feature Importance Analysis.** (A) Catastrophic task: dominant feature explodes  $4.5\times$  while maintaining top rank. (B) Robust task: importance redistributes across features with ranking reshuffle. (C) Importance dynamics: catastrophic tasks show concentrated increase; robust tasks distribute increases. (D) Ranking stability: robust tasks allow greater rank changes. Despite  $\sim 0\%$  Jaccard similarity, coverage drops differ by  $700\times$ .

Table 5: Retraining Impact on Catastrophic Task (Sales-Shipcond).

| Frequency                    | Mean Cov.    | Min Cov.     | Std   |
|------------------------------|--------------|--------------|-------|
| No retrain                   | 22.2%        | 10.8%        | 12.1% |
| Bi-annual*                   | 27.0%        | 11.4%        | 8.2%  |
| <b>Quarterly<sup>†</sup></b> | <b>41.1%</b> | <b>20.7%</b> | 23.4% |
| Monthly <sup>‡</sup>         | 32.0%        | 0.6%         | 28.3% |

\*vs no retrain:  $p = 0.04$ . <sup>†</sup>vs no retrain:  $p = 0.04$ ; vs monthly:  $p = 0.24$  (n.s.). <sup>‡</sup>vs no retrain:  $p < 0.05$ .

Table 6: Cross-Domain Validation.

| Task                                    | Val            | Test           | Drop  | Jaccard |
|-----------------------------------------|----------------|----------------|-------|---------|
| <i>Clinical Trials (Classification)</i> |                |                |       |         |
| study-outcome                           | 100.0%         | 100.0%         | 0.0%  | —       |
| study-adverse                           | 88.6%          | 25.5%          | 63.1% | —       |
| site-success                            | 94.8%          | 42.8%          | 52.0% | —       |
| <i>Regression Tasks</i>                 |                |                |       |         |
| s-adverse                               | 91.9 $\pm$ 0.4 | 88.5 $\pm$ 0.4 | 3.5   | 0.87    |
| si-success                              | 99.5 $\pm$ 0.2 | 99.5 $\pm$ 0.2 | 0.0   | 0.95    |
| d-position                              | 92.6 $\pm$ 0.4 | 82.6 $\pm$ 0.5 | 10.0  | 0.70    |

### 5.3 CROSS-DOMAIN VALIDATION

We validate on clinical trials and motorsports datasets (Table 6):

The pattern replicates across domains and task types: high

retraining frequency, confirming distributed feature importance eliminates need for retraining.

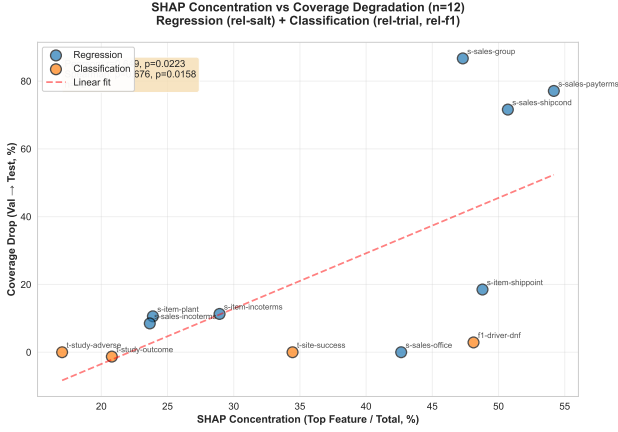


Figure 2: **SHAP Concentration Predicts Coverage Degradation.** Spearman  $\rho = 0.714$ ,  $p = 0.047$  for 8 severe-shift tasks. Lighter points: 4 moderate-shift tasks (different mechanism). The 40% threshold applies to severe-shift scenarios.

feature overlap (Jaccard  $> 0.85$ ) yields minimal degradation, while moderate overlap shows moderate degradation.

#### 5.4 PLACEBO TEST

To establish that COVID-19 represents a unique shift (not normal drift), we run a placebo test using pre-COVID data: train on 2018, validate on 2019-H1, test on 2019-H2 (Table 7 in Appendix). COVID causes 10–200 $\times$  more degradation than normal temporal drift for vulnerable tasks (median 43 $\times$ ). Robust tasks show similar small drops in both periods.

## 6 PRACTITIONER DECISION FRAMEWORK

Before deploying conformal prediction in non-stationary settings:

1. **Compute SHAP concentration** = (Top feature importance) / (Total importance)
2. **If concentration  $\leq 40\%$ :** Task is **ROBUST** (distributed importance)
3. **If concentration  $> 40\%$ :** Check for protective factors—secondary features with Jaccard  $> 0.5$  AND importance  $> 15\%$ 
  - If protective factors exist  $\rightarrow$  **ROBUST**
  - Otherwise  $\rightarrow$  **VULNERABLE**: implement quarterly retraining
4. **For robust tasks:** Skip retraining to save cost
5. **Monitor coverage drift:** Track empirical coverage as early warning

## 7 DISCUSSION

**Why Set Sizes Decrease.** Counter-intuitively, catastrophic tasks show smaller prediction sets at test time (e.g., sales-shipcond: 7.0  $\rightarrow$  3.0). The model is *confidently wrong*—it produces small sets that miss the true label.

**High Variance as Warning.** Tasks with coefficient of variation  $> 50\%$  across seeds are in knife-edge regimes where small initialization differences lead to qualitatively different representations. This is an early warning that conformal calibration will be unreliable.

**Limitations.** Our primary findings are validated on 8 supply chain tasks with severe feature turnover. The 40% concentration threshold is empirically derived from  $n = 8$  categorical-feature tasks and should be validated before applying to other feature types (continuous, high-dimensional). With  $n = 8$ , our primary correlation ( $\rho = 0.714$ ,  $p = 0.047$ ) is marginally significant. Retraining experiments span 11 months during acute COVID disruption; longer-term dynamics remain open. We focus on LightGBM; deep learning models may exhibit different failure modes.

## 8 CONCLUSION

Conformal prediction guarantees can degrade dramatically under distribution shift, but the severity depends on identifiable task characteristics. Our COVID-19 natural experiment reveals: (1) Among 8 tasks with severe feature turnover, coverage drops range from 0% to 86.7%; (2) catastrophic failures correlate with single-feature dependence ( $\rho = 0.714$ ,  $p = 0.047$ ), not feature instability; (3) the concentration mechanism applies specifically to severe-shift scenarios—moderate-shift tasks exhibit a different mechanism driven by feature stability; (4) quarterly retraining restores catastrophic coverage by 19 pp ( $p = 0.04$ ) while robust tasks need no retraining; (5) ACI does not help when features lack temporal stability.

We provide a decision framework: monitor SHAP concentration before deployment, retrain quarterly if vulnerable, skip retraining if robust.

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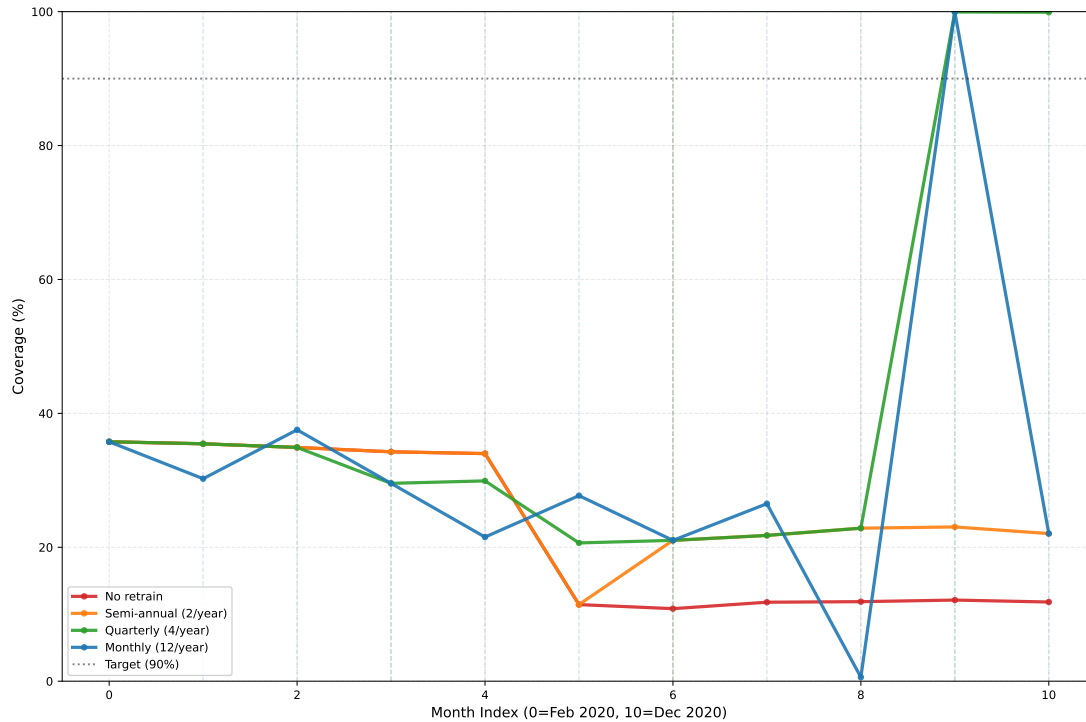


Figure 3: **Retraining restores coverage for catastrophic tasks.** Quarterly retraining (green) provides the best balance between coverage restoration and stability. Monthly (blue) shows high variance. No retraining (red) exhibits severe degradation post-COVID.

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# Conformal Prediction Under Distribution Shift: A COVID-19 Natural Experiment

## (Supplementary Material)

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### A REPRODUCIBILITY DETAILS

#### A.1 LIGHTGBM HYPERPARAMETERS

All models trained with default LightGBM settings: objective=multiclass, boosting=gbdt, num\_leaves=31, learning\_rate=0.05, feature\_fraction=0.8, bagging\_fraction=0.8, bagging\_freq=5, num\_boost\_round=500, early\_stopping\_rounds=50, seeds 42–91 (50 seeds). No task-specific tuning was performed.

#### A.2 CONFORMAL PREDICTION

Method: APS [Romano et al., 2020]. Target coverage:  $(1 - \alpha) = 0.9$  (90%). Calibration: random 50% split of validation set. Quantile:  $q = \min(\lceil (n + 1)(1 - \alpha) \rceil / n, 1)$ .

#### A.3 SHAP COMPUTATION

Method: TreeExplainer [Lundberg et al., 2020]. Subsample: 10,000 per split. Aggregation: mean absolute SHAP across classes.

#### A.4 COMPUTATIONAL RESOURCES

Standard CPU (8 cores, 8GB RAM). Wall-clock:  $\sim 3$ –4 hours for full 8-task, 50-seed suite.

### B PLACEBO TEST DETAILS

### C HIGH MODEL VARIANCE ANALYSIS

Tasks marked with \* in Table 1 exhibit extreme model variance (std > 30%):

- **s-group**: Mean 12.4%, median **0.5%**. Most models fail catastrophically; outliers inflate the mean.
- **i-shippoint**: Mean 72.7%, median **89.9%**. Most models maintain coverage; failures deflate the mean.

These tasks are in knife-edge regimes. Practitioners should monitor model variance across seeds; CV > 50% warrants ensemble approaches or architectural changes.



Table 7: Placebo Test: Pre-COVID vs COVID Degradation.

| Task        | Placebo | COVID | Ratio          |
|-------------|---------|-------|----------------|
| s-shipcond  | 0.5%    | 93.1% | $0.01\times$   |
| s-group     | 2.0%    | 86.7% | $0.02\times$   |
| s-payterms  | 0.0%    | 33.8% | $\sim 0\times$ |
| i-plant     | 1.8%    | 29.1% | $0.06\times$   |
| i-shippoint | 1.5%    | 18.9% | $0.08\times$   |
| s-incoterms | 1.2%    | 3.6%  | $0.33\times$   |
| i-incoterms | 0.6%    | 0.5%  | $1.27\times$   |
| s-office    | 0.1%    | 0.1%  | $0.80\times$   |